Smart Chair Health Monitoring System

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Project Overview

- IoT-based smart chair monitoring system
- Uses Fog and Cloud architecture
- Monitors posture and activity of office workers
- Built using AWS services and Python

Problem Statement

- Office workers sit for long hours
- Leads to poor posture and health issues
- Need real-time monitoring and alerts
- IoT + Fog Computing helps reduce latency

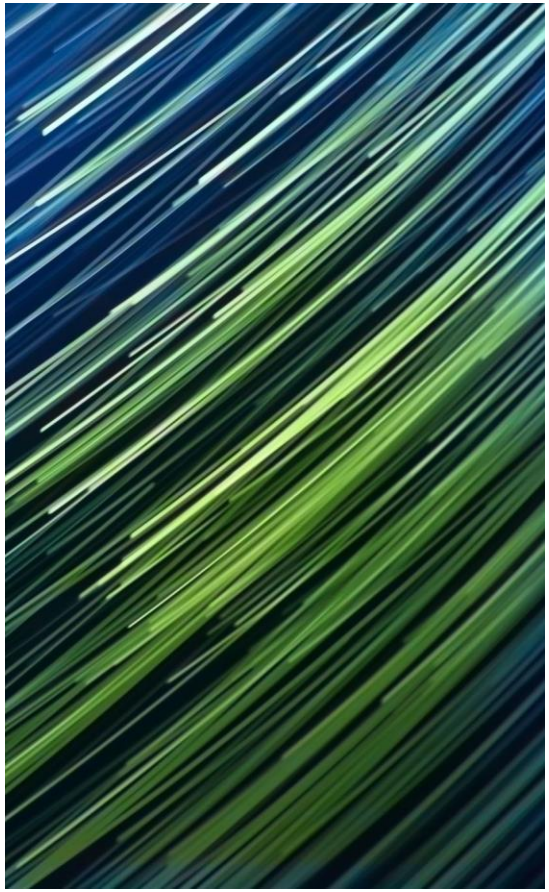


System Architecture

Sensor Simulator → Fog Node (EC2) → API Gateway → Lambda → SQS → Lambda → DynamoDB → Dashboard

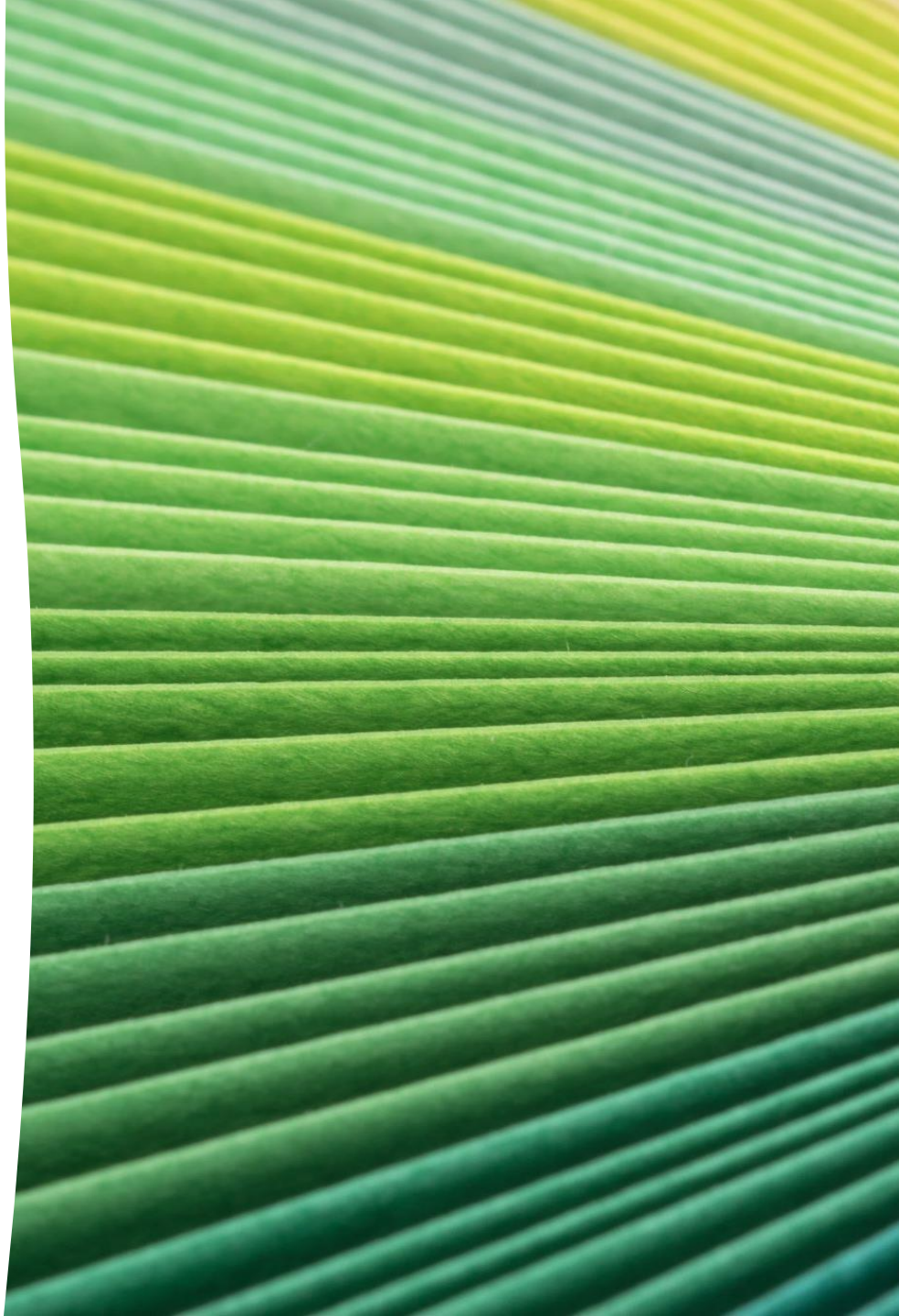
Sensor Types Used

- Seat Pressure Sensor
- Posture Angle Sensor
- Heart Rate Sensor
- Temperature Sensor
- Movement Detection



AWS Services Used

- EC2 (Fog Node)
- API Gateway
- AWS Lambda
- SQS Queue
- DynamoDB
- Terraform (Infrastructure as Code)





Data Flow

- Sensor simulator generates health data
- Data sent to EC2 fog node
- Fog node processes and sends alerts
- Data forwarded to API Gateway
- Lambda sends message to SQS
- Second Lambda stores data in DynamoDB
- Dashboard visualizes sensor data

Dashboard Features

- Live sensor readings
- Graphs using Chart.js
- Posture alerts
- Movement monitoring
- Real-time data visualization

Results


• Real-time monitoring achieved
• Alerts for bad posture and inactivity
• Data successfully stored in DynamoDB
• Scalable fog-to-cloud architecture implemented

Conclusion & Future Work

- Demonstrates IoT monitoring using fog computing
- Cloud backend enables scalability

Future Improvements:

- Real hardware sensors
- Mobile notifications
- Machine learning posture prediction
- Multiple chair monitoring

